

Image Acquisition Noise and Its Minimization

Question: Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Theory

Image acquisition noise originates from sensor electronics, thermal effects, photon shot noise, analog-to-digital conversion and low-light conditions. These factors reduce image quality by introducing random intensity variations.

Noise Reduction

Noise can be minimized using better sensors, proper illumination, suitable camera settings and image processing filters such as Gaussian, Median and Bilateral filters. The provided OpenCV program demonstrates adding Gaussian sensor noise and reducing it using these filters.

Python Code

See attached Colab code from the conversation (upload image, add Gaussian noise, apply Gaussian/Median filters)

Result

The denoised images show improved visual quality compared with the noisy image, illustrating how post-processing improves acquired sensor images.